

Some thoughts about reducing Boolean expressions

NOTE: it is important to read the notes (section 1) before reading this document. This document provides some additional thoughts about reducing Boolean expressions.

The link between truth tables, circuits and Boolean expressions

It is important to know the following (see the notes):

- how to create a truth table
- how to create a Boolean expression from a truth table
- how to reduce a Boolean expression
- how to create a circuit from a Boolean expression

You should be able to use this process from any point, e.g. start with a circuit and produce a truth table.

How useful is it to reduce Boolean expressions?

Reducing Boolean expressions is very important for people who actually design real circuits. A reduced expression will lead to a circuit with less gates, thus saving space and money.

HOWEVER, 159.102 is a Computer Science course, not an electronics course. We need to recognise that reducing Boolean expressions is important but it is not a major topic in our course.

Ok, so what should we know about reducing Boolean expressions?

In theory, you should be ready to use any of the axioms (rules) while reducing a Boolean expression. In practice (in 159.102) we mainly use only one approach, demonstrated in the following example:

Example: reduce the following Boolean expression:

$$A \bullet \sim B \bullet C + A \bullet \sim B \bullet \sim C + \sim A \bullet B \bullet C$$

Note that this Boolean expression is typical of what you get from a truth table.

Use the axiom: $P \bullet Q + P \bullet R = P \bullet (Q + R)$ where $P = (A \bullet \sim B)$, $Q = C$ and $R = \sim C$

So we get: $(A \bullet \sim B) \bullet (C + \sim C) + \sim A \bullet B \bullet C$

Now use the axiom: $P + \sim P = \text{TRUE}$

So we get: $(A \bullet \sim B) \bullet (\text{TRUE}) + \sim A \bullet B \bullet C$

Now use the axiom: $\text{TRUE} \bullet P = P$

So we get: $(A \bullet \sim B) + \sim A \bullet B \bullet C$

This is a significant reduction in the number of gates required for the final circuit. You can check your solution by designing a circuit for the original (non-reduced) expression and also designing a circuit for the reduced expression. The two circuits should produce the same truth table. If you set up a simulation for the two circuits, you should get the same results (light on or off) for the same settings of the switches.

This type of reduction covers almost all of what we need to know for 159.102. We are not interested in reducing every type of Boolean expression, we are interested in reducing circuits that are created from truth tables.

[Questions in a quiz will almost certainly be about this type of example. Is this a hint...?]

Note that the reduction process can be stopped at any point. In this example, we could stop at any of the following steps and create a circuit:

$A \bullet \sim B \bullet C + A \bullet \sim B \bullet \sim C + \sim A \bullet B \bullet C$
the original expression (not reduced)

$(A \bullet \sim B) \bullet (C + \sim C) + \sim A \bullet B \bullet C$
a partially reduced expression

$(A \bullet \sim B) + \sim A \bullet B \bullet C$
a fully reduced expression (or maybe it can be reduced further...)

You could build a circuit from any one of these expressions and the circuits would all do the same thing (generate the same truth table). The only difference is that they have a different number of gates.

A quiz question may say “reduce the expression and create a circuit”. You could stop your reduction at any point and create a circuit – and it would be a correct circuit. You may lose 1 or 2 marks (out of 10) if you stopped the reduction early.

Please note that these are some extra thoughts about reducing Boolean expressions. They are not intended to replace the section about reducing Boolean expressions in the notes.